

BATAM HANG NADIM, IndonesiaWMO: **960870**Lat: **1.1192N**Lon: **104.1136E**Elev: **24**StdP: **101.04**Time Zone: **7.00 (E07)**Period: **02-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	23.8	20.2	14.9	27.8	20.9	15.6	27.8	10.5	29.7	10.0	29.7	1.7	250	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.8	32.7	25.8	32.2	25.9	31.9	25.9	27.2	30.4	26.9	30.1	26.7	29.8	4.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.7	28.5	26.1	21.6	28.4	25.9	21.3	28.2	85.8	30.8	84.6	30.4	83.8	30.1	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	8.0	7.0	21.6	34.3	1.9	1.1	20.2	35.1	19.1	35.7	18.0	36.3	16.6	37.1
			20.7	28.0	1.6	0.6	19.5	28.4	18.6	28.8	17.7	29.2	16.5	29.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	27.3	27.6	28.1	28.4	28.4	28.3	27.9	27.9	27.8	27.9	27.5	27.3
	DBStd	1.09	1.19	0.97	1.10	0.98	0.93	1.09	1.10	1.11	1.01	0.92	0.88	1.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6523	536	493	561	551	569	548	556	556	534	556	526	538
	CDD18.3	3481	278	260	302	301	311	298	297	297	284	297	276	279
	CDH23.3	36459	2666	2654	3293	3329	3436	3268	3168	3161	2957	3077	2744	2706
CDH26.7	11299	668	731	1084	1166	1179	1120	1019	999	935	967	754	677	
Wind	WSAvg	2.6	3.8	4.2	3.3	2.1	1.9	2.1	2.6	2.7	2.3	1.7	1.7	2.7
Precipitation	PrecAvg	2418	247	111	182	190	185	161	173	184	170	216	281	325
	PrecMax	3068	647	328	421	316	346	265	253	401	271	379	462	919
	PrecMin	1436	6	16	39	92	56	86	60	96	47	60	131	123
	PrecStd	434	207	90	105	63	67	61	50	69	52	95	94	183
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	31.7	32.5	33.2	33.1	33.0	32.2	32.2	32.3	33.0	32.2	31.9
		MCWB	25.6	25.3	25.6	26.0	26.2	25.5	25.9	25.6	25.5	24.9	25.7	25.5
	2%	DB	31.0	31.1	31.8	32.3	32.3	32.2	31.7	31.7	31.8	32.1	31.4	31.0
		MCWB	25.5	25.3	25.6	26.1	26.3	26.1	25.9	25.6	25.6	25.4	25.7	25.5
	5%	DB	30.2	30.6	31.2	31.8	31.8	31.7	31.2	31.1	31.2	31.3	30.9	30.3
		MCWB	25.2	25.1	25.5	26.1	26.3	26.2	25.9	25.7	25.6	25.6	25.7	25.4
	10%	DB	29.8	30.0	30.7	31.1	31.1	31.1	30.7	30.6	30.8	30.7	30.1	29.7
		MCWB	25.0	24.9	25.5	26.0	26.3	26.2	25.8	25.6	25.6	25.6	25.6	25.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.3	26.7	27.2	27.6	27.3	27.1	26.7	26.8	26.7	26.7	26.6
		MCDB	29.9	29.8	30.5	30.7	30.8	30.7	30.3	29.9	30.1	29.8	29.7	29.6
	2%	WB	26.1	26.0	26.2	26.9	27.2	27.0	26.7	26.5	26.4	26.4	26.3	26.2
		MCDB	29.2	29.5	29.9	30.3	30.3	30.4	29.9	29.6	29.7	29.5	29.2	29.1
	5%	WB	25.8	25.7	26.1	26.6	26.8	26.7	26.4	26.2	26.2	26.2	26.1	25.9
		MCDB	28.7	29.0	29.6	29.9	29.8	29.9	29.5	29.3	29.4	29.2	29.0	28.8
	10%	WB	25.5	25.4	25.8	26.4	26.6	26.5	26.2	26.0	25.9	26.0	25.9	25.7
		MCDB	28.4	28.6	29.2	29.5	29.5	29.5	29.2	29.0	29.1	29.0	28.7	28.4
Mean Daily Temperature Range	5% DB	MDBR	4.7	4.9	5.3	5.8	5.8	5.7	5.4	5.4	5.8	5.9	5.4	4.8
		MCDBR	5.4	5.4	5.7	6.5	6.4	6.4	6.0	5.8	6.5	6.7	6.4	5.6
		MCWBR	1.7	1.6	1.6	2.0	2.1	2.0	1.9	1.9	2.1	2.1	2.0	1.8
	5% WB	MCDBR	5.1	5.1	5.5	6.1	5.9	5.9	5.6	5.5	6.1	6.1	5.7	5.2
		MCWBR	1.8	1.7	1.8	2.2	2.2	2.2	2.1	2.1	2.3	2.2	2.1	1.9
Clear-Sky Solar Irradiance	taub	0.470	0.481	0.486	0.475	0.463	0.473	0.467	0.510	0.543	0.571	0.480	0.470	
	taud	2.308	2.260	2.253	2.284	2.331	2.304	2.327	2.181	2.082	1.985	2.273	2.316	
	Ebn at Noon	862	857	847	837	826	804	815	798	788	772	846	855	
	Edh at Noon	135	144	145	137	126	127	126	150	169	187	139	133	
All-Sky Solar Radiation	RadAvg	4.58	5.32	5.28	5.08	4.69	4.44	4.42	4.53	4.54	4.48	4.18	4.11	
	RadStd	0.38	0.47	0.45	0.32	0.23	0.32	0.36	0.28	0.29	0.33	0.28	0.32	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.37	+0.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page